

ADV7403/ADV7401 Multiformat Video Decoders for Advanced TV

Key Features

- 12-bit NSV® (noise shaped video)
- True 10-bit video data processing and output
- 140 MSPS conversion rate supports RGB input resolutions up to $1,280 \times 1,024 @ 75 \text{ Hz}$ (SXGA)
- NTSC, PAL, and SECAM color standards support
- Chroma transient improvement (CTI) and digital noise reduction (DNR) for SDTV formats
- Oversampled fast switch overlay support for SCART and OSD mixing applications
- Integrated antialiasing filters
- Full component HDTV input support for 525p, 625p, 720p, and 1080i, plus many other HD and SMPTE standards
- CVBS, S-Video, YPrPb, and RGB input support, including 5-wire YPrPbHV/RGBHV and CSync
- 4× oversampling (54 MHz) for standard definition and 2× oversampling for progressive scan inputs
- 5-line, super adaptive comb filters for advanced NTSC and PAL 2D Y/C separation
- 24-bit, digital RGB, input port accepts data from common DVI/HDMI receivers
- Any-to-any, 3×3 , color space converter
- Fully certified Macrovision® detection for 525i, 625i, 525p, and 625p
- ADLLT™ (adaptive digital line length tracking) enables robust sync stability for poor video sources such as VCRs and camcorders, including noisy, weak, or distorted composite video signals from tuners and other nonstandard sources
- Digital automatic gain control (AGC) for super fast response-to-step input changes
- New VBI data slicer with Teletext support, also supports CGMS, WSS, EDTV, Gemstar™ 1×/2×, CC, and XDS

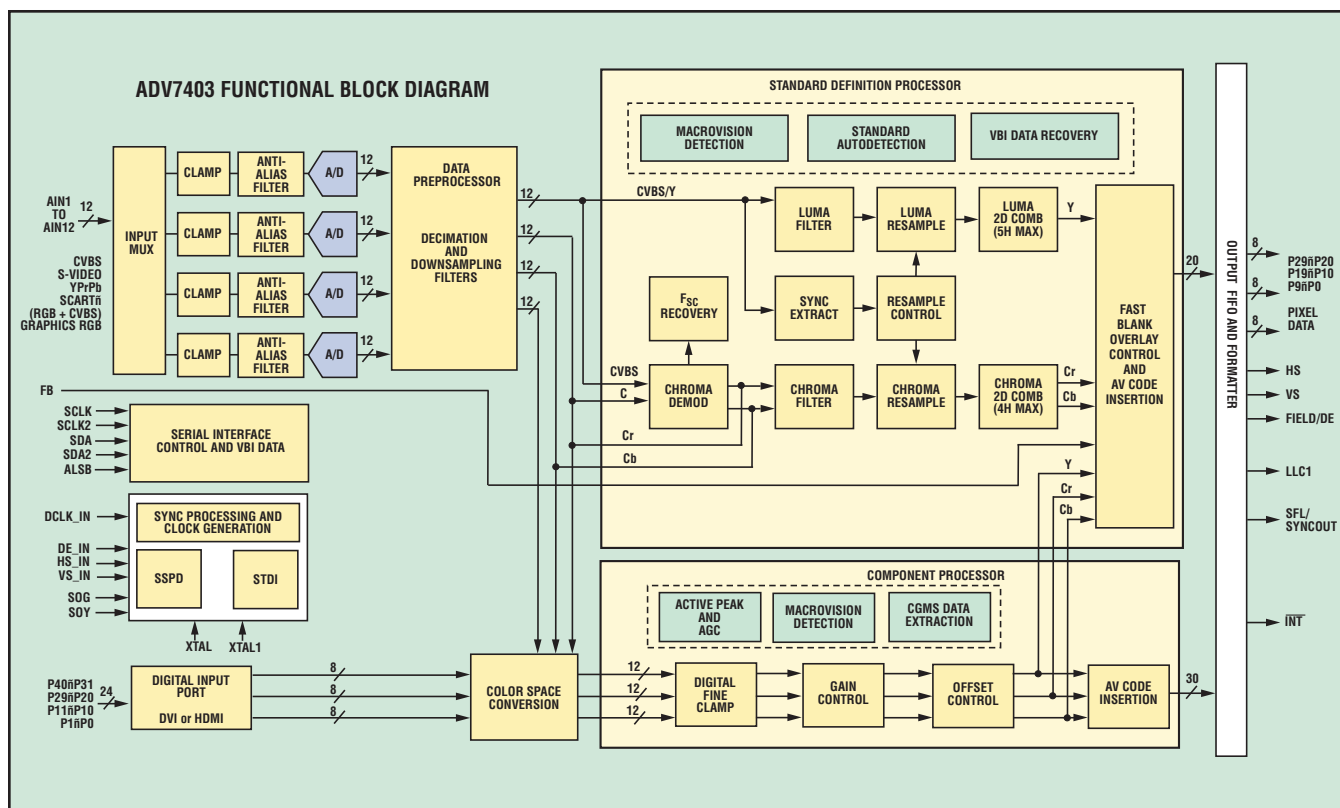


Integrated Multiformat SDTV/HDTV Video Decoder and RGB Graphics Digitizer

The ADV7403 high performance multiformat video decoder is the first to integrate 12-bit quality video decoding with high speed graphics and high definition video digitization. Designed for the highest mixed-signal performance, noise shaped video ADCs enable true 10-bit data output accuracy for all sources up to HDTV rates.

With 12 analog input channels, the ADV7403 enables system designers to choose one part to route or convert multiple analog video inputs into industry-standard or proprietary output interfaces. With support for single data rate (SDR) and double data rate (DDR) output modes, interfacing to popular scalars, MPEG A/V codecs, and graphics processors is easily achieved.

Using the on-board, 3×3 conversion matrix, digital, or analog RGB inputs can be converted to any color space. ADV7403 is suitable for a wide range of domestic and worldwide consumer electronics designs, and is ideal for home theater projectors or direct view and rear projection HDTVs, where two or more chips (decoder and ADC) can be replaced with one ADV7403 IC. For cost-sensitive applications, the ADV7401 provides designers with high performance 10-bit ADCs and an 8-bit accurate output interface. Now with full support for SCART inputs, teletext data slicing, and integrated antialiasing filters on each ADC channel, the ADV7403 and ADV7401 have been fully optimized for TV applications.



ADV7403 Product Specifications

- 12-bit ADCs: 65 dB SNR (luma flat field)
- 0.4% differential gain, 0.4° differential phase (typical)
- 100-lead, 100% Pb-free 14 mm × 14 mm LQFP
- -40°C to +85°C temperature ranges*

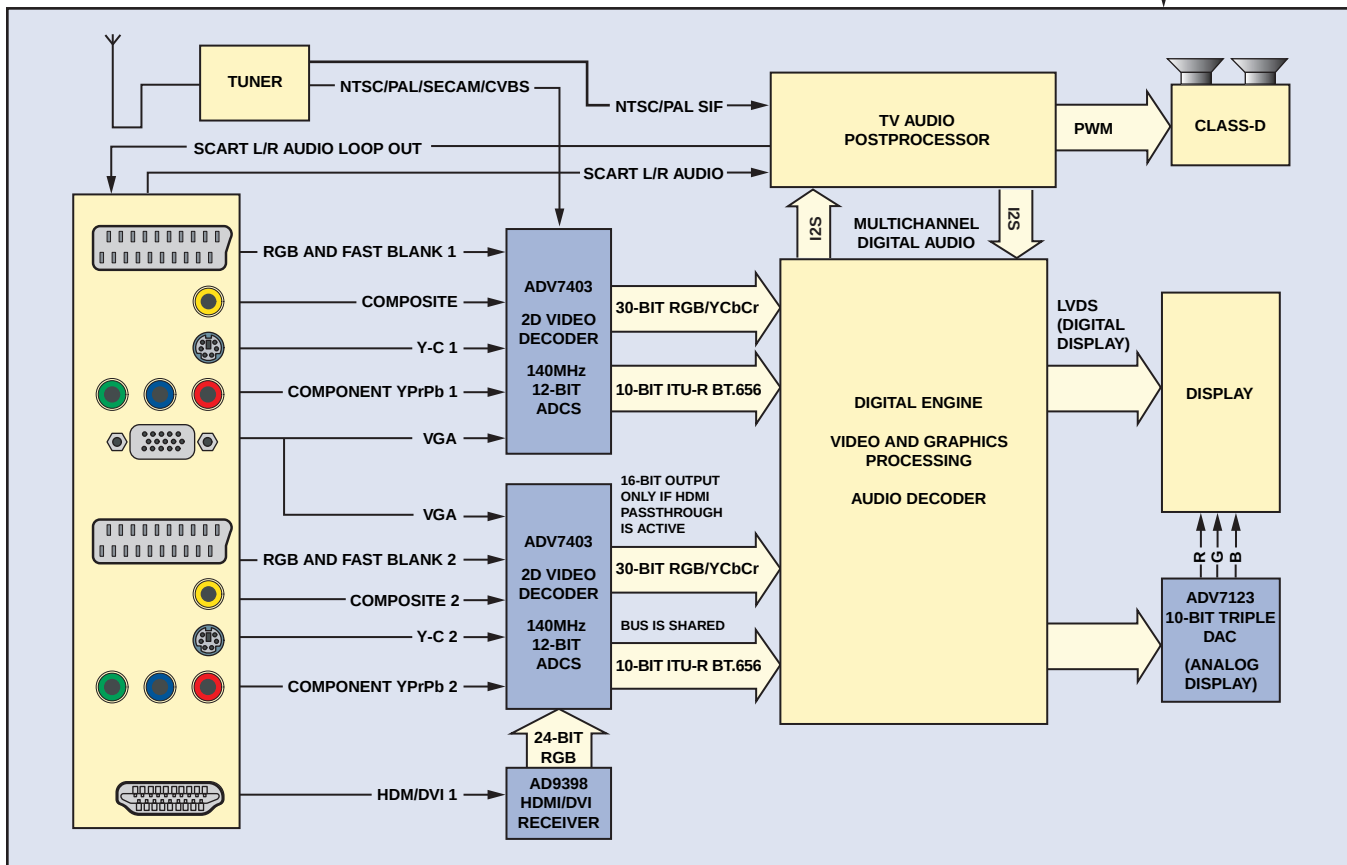
*140 MHz speed grades 0°C to +70°C only.

ADV7403 Product Specifications

- 0.5 V to 1.6 V analog signal input range
- 3.3 V analog, 1.8 V digital core, 3.3 V I/O power supplies
- 2-wire serial MPU interface, I²C®-compatible (400 kbps max)
- 30-bit configurable pixel output port
 - 8-bit/10-bit 4:2:2 YCbCr
 - 16-bit/20-bit 4:2:2 YCbCr
 - 24-bit/30-bit 4:4:4 YCbCr/RGB
- Double-data-rate (DDR) output modes
 - 8-bit 4:2:2 YCbCr
 - 12-bit 4:4:4 RGB (graphics)
- 10-bit digital baseband composite video input for external tuner support
- Automatic VCR/TV detection and processing
- VBI data output through pixel port interface (ancillary data packets as per ITU-R BT.1364) for teletext, closed captioning (CC), and Gemstar™ 1×/2×
- Other VBI data services accessible via the I²C interface include copy generation management system (CGMS), wide screen signalling (WSS), EDTV, closed captioning, and extended data services (XDS)
- Support for all worldwide color formats:
 - PAL B/D/I/G/H, PAL60, PAL-M, PAL-N, PAL-Nc
 - NTSC M/J, NTSC 4.43
 - SECAM B/D/G/K/L
 - NTSC square pixel support

Analog Devices offers a diverse range of products to complete the advanced TV signal chain, including multiformat video decoders (ADV7403), dual interface graphics and HDMI/DVI receiver products (AD9880¹), video encoders (ADV7320), and video DACs (ADV7123). Standard definition video decoders, including the ADV7183B, ADV7189B, and ADV7180 are also available for multichannel configurations and other video recording and processing applications. The AD1991 Class-D audio power amplifier output stage also enables state-of-the-art, compact electronic component design for home theater systems and flat panel displays.

¹Send email to flatpanel_apps@analog.com for more detailed information.



The ADV7403 and ADV7401 multiformat video decoders simplify front end digitizer and decoder architectures by reducing board space, simplifying autodetection and switching of inputs, and reducing overall system power consumption.

For TV configurations, the ADV7403/ADV7401 can process and switch between the following signal types without the need for external multiplexer circuitry:

- RGB graphics outputs from a PC graphics card
- YPrPb outputs from a progressive scan DVD player or high definition set-top box
- Composite video from a TV tuner or VCR
- S-Video from a camcorder or S-VHS VCR

Ideally suited to applications that require support for all these inputs, the ADV7403/ADV7401 can interface directly to many graphics controllers, scalars, deinterlacers, MPEG encoders, and image enhancement processors with bus widths as low as 10 bits (+clock) for all scanning rates and sampling schemes.

Evaluation kits for the ADV7403 and ADV7401 are now available from Analog Devices. Each kit includes the equipment necessary to start prototyping with those devices. With the on-board ADuC831 microcontroller, each kit comes ready programmed for standalone operation, including preprogrammed autodetection and mode selection via the on-board rotary hex switches.

Each video decoder module is equipped with standard RCA/phono type connectors for component and composite video inputs. A mini-Din connector is also provided for S-Video connections, along with a 15-lead D-Sub for standard RGB graphics inputs.

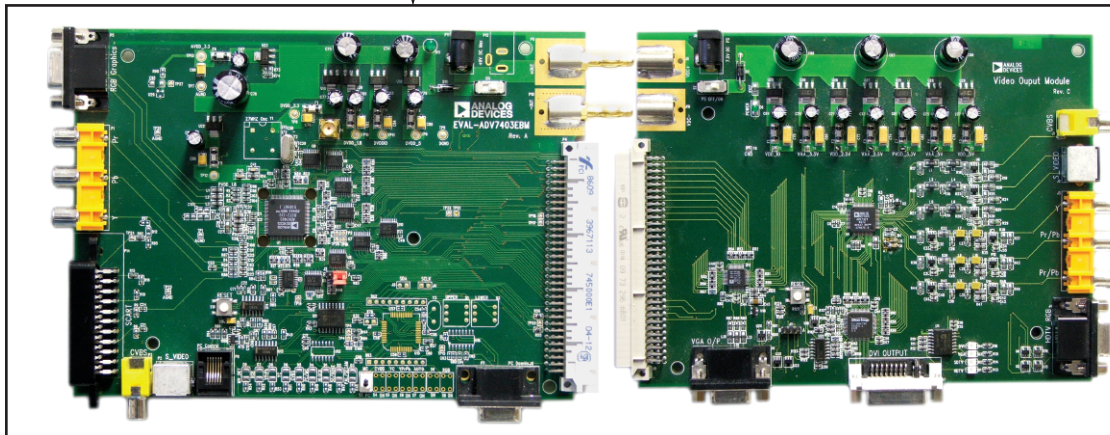
A display output module equipped with video encoder (ADV7321), graphics DAC (ADV7123), and DVI transmitter provides a variety of output formats to interface with professional analog or digital displays and basic televisions. Control software is provided for complete configuration of each module via the parallel port of a PC using serial communications. Various add-on modules are also available to provide data bus expansion and support for future functionality.

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Ordering Information

Part Number ¹	Description
ADV7403BSTZ-110 ²	–40°C to +85°C temperature range, 100-lead, low profile quad flat package (LQFP)
ADV7403KSTZ-140 ²	0°C to +70°C temperature range, 100-lead, low profile quad flat package (LQFP)
ADV7401BSTZ-80 ²	–40°C to +85°C temperature range, 100-lead, low profile quad flat package (LQFP)
ADV7401BSTZ-110 ²	–40°C to +85°C temperature range, 100-lead, low profile quad flat package (LQFP)
ADV7401KSTZ-140 ²	0°C to +70°C temperature range, 100-lead, low profile quad flat package (LQFP)
EVAL-ADV7403EBM	Evaluation board
EVAL-ADV7401EBM	Evaluation board

¹The ADV7403 is a Pb-free, environmentally friendly product. It is manufactured using the most up-to-date materials and processes. The coating on the leads of each device is 100% pure Sn electroplate. The device is suitable for Pb-free applications, and is able to withstand surface-mount soldering at up to 255°C (±5°C). In addition, it is backward-compatible with conventional SnPb soldering processes. This means that the electroplated Sn coating can be soldered with SnPb solder pastes at conventional reflow temperatures of 220°C to 235°C.

²Z = Pb-free part.

For ADV7403 and ADV7401 samples, evaluation boards, or availability information, please contact us via email at AV.products@analog.com.

For more information on these products and Analog Devices' extensive video and imaging product portfolio, please visit our website.

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